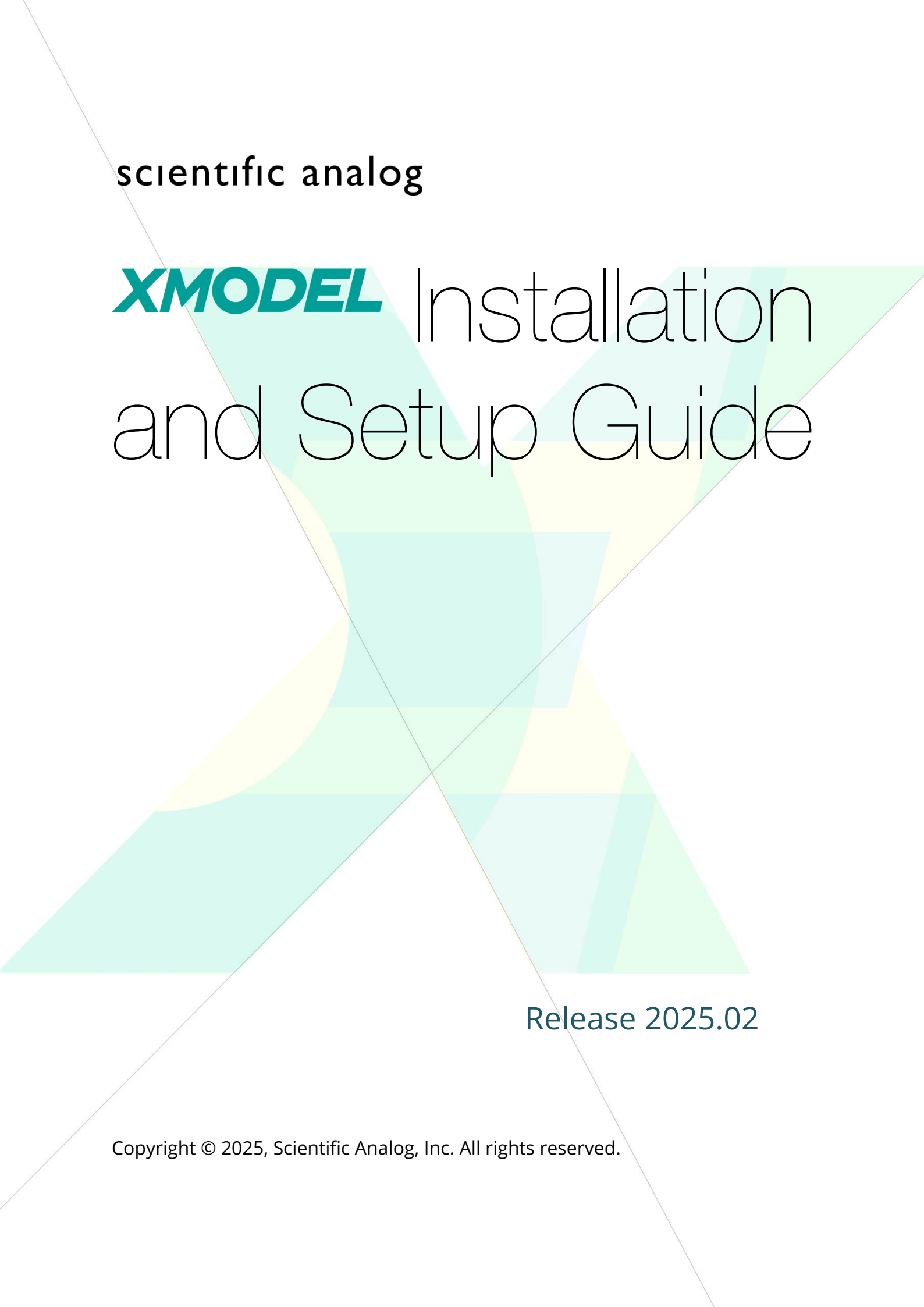


scientific analog



XMODEL Installation and Setup Guide

Release 2025.02

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1 About This Document

The **XMODEL** Installation Guide provides the basic information and procedures required to install **XMODEL** software and its supporting utilities.

2 Setting Up FlexNet Publisher License Server (R2023)

Scientific Analog, Inc. offers two types of licensing models: floating (concurrent) licensing and node-locked licensing. And its products, **XMODEL**, **GLISTER**, and **MODELZEN** use FlexNet Publisher from Flexera Software, LLC to administer the usages of their licensed features. This guide describes a step-by-step procedure to install and setup the FlexNet Publisher license server on your network for floating licensing. Note that a license server is not required for node-locked licensing.

Installing and setting up the FlexNet Publisher consists of five simple steps:

1. Choose a License Server Machine
2. Collect Information on the License Server Machine
3. Install the License Server Components
4. Install the License File and Start the License Manager
5. Set up End-user Client Machines to Access the License Server

For detailed information on how to use FlexNet Publisher, please refer to the *FlexNet Publisher License Administration Guide* included in the installation package or on our support website, <http://www.scianalog.com/support/>.

2.1 Choose a License Server Machine

First of all, choose a machine that will act as a license server. A stable machine that does not get frequently rebooted or shut down is desirable. The license server machine does not need to have the same architecture or operating system as the end-user client machines where **XMODEL** are running. For further information and tips on how to choose the license server machines, please refer to the “Selecting a License Server Machine” and “Ensuring License Availability” sections in the *FlexNet Publisher License Administration Guide*.

2.2 Collect Information on the License Server Machine

The license files issued by Scientific Analog, Inc. are locked to a specific license server machine you choose. On Linux platforms, the FlexNet Publisher uses two information fields to identify a server machine: the *hostname* and *Ethernet MAC address*. Hostname is a simple string (e.g. **'mylicserver'**) that identifies a machine and Ethernet MAC address is six groups of two hexadecimal digits (e.g. **00:1C:42:3C:3F:21**) that uniquely identifies each network interface connected to the network.

Basically, you need to supply the hostname and Ethernet MAC address of your license server machine to Scientific Analog, Inc. to get the license file issued.

One way to figure out the hostname and Ethernet MAC address of your chosen license server machine is to use the FlexNet utility called **lmhostid**, which you can download from the Flexera Software website or install by performing the Step 3 described in

the next section. Once you have this **lmhostid** installed, you can retrieve the hostname and Ethernet MAC address information by executing the following commands on your license server machine:

```
unix> lmhostid -hostname
lmhostid - Copyright (c) 1989-2013 Flexera Software LLC. All Rights Reserved.
The FlexNet host ID of this machine is "HOSTNAME=mylicserver"

unix> lmhostid -ether
lmhostid - Copyright (c) 1989-2013 Flexera Software LLC. All Rights Reserved.
The FlexNet host ID of this machine is "001c423c3f21"
```

Alternatively, you can find out the hostname and Ethernet MAC address of your license server machine using the more basic Linux commands. For instance, the following commands return the hostname of your machine and Ethernet MAC addresses of the network interfaces installed on the machine:

```
unix> hostname
mylicserver

unix> /sbin/ifconfig -a
eth0      Link encap:Ethernet  HWaddr 00:1C:42:3C:3F:21
          inet addr:10.111.55.3  Bcast:10.111.55.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:15890 errors:0 dropped:0 overruns:0 frame:0
          TX packets:15076 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1226510 (1.1 MiB)  TX bytes:44425220 (42.3 MiB)
```

In the above case, the **HWaddr** field indicates the Ethernet MAC address of the interface **eth0**. Note that your machine may give different outputs depending on its configurations and platforms. In case you have multiple network interfaces installed on the server machine, supply the Ethernet address of the network interface through which the license server is connected to the end-user client machines. When it is not certain which one is the right one, let the **lmhostid** utility decide it for you.

The FlexNet Publisher license server also supports three-server redundant triad for higher stability and availability. If you wish to setup the license server as three-server redundant triad, supply the hostnames and Ethernet MAC addresses of the three license server machines to Scientific Analog, Inc.

2.3 Install the License Server Components

The next step is to install the FlexNet license server components. Namely, they are the FlexNet license manager (**lmgrd**), Scientific Analog licensing daemon (**salmd**), and FlexNet utilities (**lmutil**).

You can install these license server components by downloading the Scientific Analog's FlexNet License Manager Installer from <http://www.scianalog.com/support/downloads/license-manager> to your license server machine and executing the following commands:

```
unix> chmod +x install_sa_licmgr_R2023.sh
unix> ./install_sa_licmgr_R2023.sh
```

Then the installer will prompt you for the path where you would like to install the license server components and operating system of the license server machine (**x86_64** for 64-bit Linux and **x86** for 32-bit Linux). For instance:

```
unix> ./install_sa_licmgr_R2023.sh

Welcome to the Scientific Analog License Manager Installer!

*** Choose the installation path (press enter to accept the default):
[/cad/downloads] >>> /cad/scianalog/licmgr

*** Choose the OS architecture for the binary files [x86_64|x86]:
[x86_64] >>> x86_64
```

You can also install the license server components in a batch mode without the interactive prompts by using the **-b** option and specifying the installation path and operating system with the **-p** option and **-r** option, respectively. For instance:

```
unix> ./install_sa_licmgr_R2023.sh -b -p /cad/scianalog/licmgr -r x86_64
```

To see the full descriptions on the available options and their default values, give the **-h** option to the installer command:

```
unix> ./install_sa_licmgr_R2023.sh -h
usage: install_sa_licmgr_R2023.sh [options]

Installs the FlexNet Publisher license server components for
Scientific Analog products, by default to the path:
/cad/downloads

-h          print this message and exit
-b          run in batch mode (non-interactive)
-r <arch>   OS architecture (x86_64 or x86, default: x86_64)
-l <file>    license file (default: None)
-p <path>    change the installation path (overriding the default)
```

2.4 Install the License File and Start the License Manager

When the license manager installer successfully completes its operation, it displays messages on how to install the license file and start/stop the license manager. For instance:

```
-----
Congratulations!
The Scientific Analog License Manager is successfully installed.
-----

*** You can now start/stop the license server using the following commands:

To start the license server:
/cad/scianalog/licmgr/bin/lmgrd -c /cad/scianalog/licmgr/license/xmodel.lic

To stop the license server:
/cad/scianalog/licmgr/bin/lmdown -c /cad/scianalog/licmgr/license/xmodel.lic

Advanced users can include these commands in the proper start-up scripts
(e.g. /etc/rc.local) to automatically start/stop the license server at the
system boot/shutdown.

*** The end-user machines should set either LM_LICENSE_FILE or
SALMD_LICENSE_FILE environment variable to define the license server name and
port information. For example:

For bash-like shells:
export SALMD_LICENSE_FILE=9524@boazRH5

For csh-like shells:
setenv SALMD_LICENSE_FILE 9524@boazRH5

*** Note that a license file could not be installed as it was not specified or
found. If you need to request a new license file, please supply the following
information to Scientific Analog:

-----
The FlexNet host ID of this machine is "HOSTNAME=mylicserv"
The FlexNet host ID of this machine is "001c423c3f21"
-----

When you receive a new license file, copy it to
/cad/scianalog/licmgr/license/xmodel.lic and restart the license server.

Thank you for using XMODEL!
```

The recommended location to store the license file is in the subdirectory named **license** under the license manager installation directory. In fact, if you supply the license file during the installation process (e.g. by specifying the **-l** option), the license file will be copied into the path of **license/xmodel.lic**. The following description will assume that the license file is located there.

Once the license file is installed on the license server machine, you can start the license manager by executing the following command:

```
unix> $LM_INST_PATH/bin/lmgrd -c $LM_INST_PATH/license/xmodel.lic
```

where **\$LM_INST_PATH** denotes the installation path of the license manager. If you have installed the license file in other places than **\$LM_INST_PATH/license/xmodel.lic**, you can specify its full path instead after the **-c** option. It is recommended that the license manager not be started by a superuser (i.e. root). A user account with the proper access privileges to the license server component directories is sufficient to start the license server. The command **lmgrd** has more available options (e.g. for specifying a debug log file). For more details, please refer to the *FlexNet Publisher License Administration Guide*.

On the other hand, you can stop by the license manager by executing the following command:

```
unix> $LM_INST_PATH/bin/lmdown -c $LM_INST_PATH/license/xmodel.lic
```

An advanced user can setup the license server machine so that the license manager is automatically started at the system boot and stopped at shutdown. For instance, you can insert the start command in the **/etc/rc.local** file. For more information, please

refer to the UNIX administration guides suitable for your operating system.

The FlexNet Publisher comes with a variety of command-line utilities to retrieve the information regarding the license server running.

Table 1 lists the available commands. For detailed information on how to use each of these commands, please refer to the *FlexNet Publisher License Administration Guide*.

Table 1. Command-line utilities for FlexNet Publisher license administration.

Utility Name	Description
Imborrow	Supports license borrowing.
Imdiag	Diagnoses license checkout problems.
Imdown	Gracefully shuts down selected vendor daemons on the license server (or on all three systems in the case of three-server redundancy).
Imhostid	Reports the hosted of a system.
Iminstall	Converts license files between different formats.
Imnewlog	Moves existing report log information to a new file name and starts a new report log file with the existing file name.
Impath	Allows users direct control over license file path settings.
Imremove	Releases a hung license to the pool of free licenses.
Imreread	Causes the license daemon to reread the license file and start any new vendor daemons.
Imstat	Displays the status of a license server.
Imswitch	Controls debug log location and size.
Imswitchr	Switches the report log to a new file name.
Imver	Reports the version of a library or binary file.
Imvminfo	Returns the environment of the system used as to virtual or not.

2.5 Set up End-user Client Machines to Access the License Server

An end-user client machine needs to access the license server machine on the network in order to check out the desired license features. Define the environment variable named **SALMD_LICENSE_FILE** to specify the location of the server machine and its service TCP port number (by default, 9524).

On a bash-like shell:

```
unix> export SALMD_LICENSE_FILE=9524@<SERVER>
```

On a csh-like shell:

```
unix> setenv SALMD_LICENSE_FILE 9524@<SERVER>
```

where **<SERVER>** denotes the hostname of the license server machine. It is convenient to define an entry in the **/etc/hosts** file in case the mapping from this hostname to the IP address is not defined elsewhere. Also, to avoid having to define the environment variable every time a new shell is created, add the above command to the initialization script such as **.bashrc** or **.cshrc**.

In case you have multiple license servers, you can specify all of them using colons (':') or commas(',') as delimiters. For instance:

```
unix> export SALMD_LICENSE_FILE=9524@<SERVER1>:9524@<SERVER2>:9524@<SERVER3>
```

or,

```
unix> setenv SALMD_LICENSE_FILE 9524@<SERVER1>,9524@<SERVER2>,9524@<SERVER3>
```

where <SERVER1>, <SERVER2>, and <SERVER3> are the hostnames of the license server machines.

3 Installing and Setting Up XMODEL

This page describes the step-by-step procedure to install the **XMODEL** package and setup the environment variables on your system.

3.1 Installing the XMODEL Package

The **XMODEL** package is distributed as a single self-extraction shell-script and you can download the latest release from [this page](#).

Currently, **XMODEL** supports *32-bit and 64-bit RedHat Enterprise Linux (RHEL) version 5 or later*. The name of the installation script reflects the version of the **XMODEL** release and intended OS platform. For instance, the installation script of the 2025.02 Release for 64-bit Linux is named: **xmodel_2025.02_x86_64.sh**.

Install the **XMODEL** package is easy – all you need is to execute the script on the shell prompt.

Note that depending on how you downloaded the installation script, you may have to make the script executable first by using the following command:

```
chmod +x xmodel_2025.02_x86_64.sh
```

Then, for an interactive-mode installation, type:

```
./xmodel_2025.02_x86_64.sh
```

The script will ask for the installation path, prompts for licensing agreement, etc. and guide the installation process in a friendly way.

On the other hand, for a batch-mode installation (i.e. installing the package in a single step with no questions asked), you can use the following options with the script:

```
./xmodel_2025.02_x86_64.sh -b -p <XMODEL_INSTALLATION_PATH>
```

Fill the field **<XMODEL_INSTALLATION_PATH>** with the directory path where you want to install **XMODEL**. The script will then install the package to your designated installation path without prompting the user.

3.2 Setting Up the XMODEL Environments

To run **XMODEL**, users need to set the following three or four environment variables:

- **\$XMODEL_HOME** variable to indicate where **XMODEL** package is installed.
- **\$PATH** variable appended with the **\$XMODEL_HOME/bin** executable directory.
- **\$SALMD_LICENSE_FILE** variable defining the license server information.
- (Optional) **\$XMODEL_SIMULATOR** variable to specify which Verilog simulator to use (Synopsys® VCS®, MentorGraphics® ModelSim®, or Cadence® NC-Verilog®) with **XMODEL**.

For bash-like shells, you can use the following commands (see **\$XMODEL_HOME/etc/DOT.bashrc** for example):

```
export XMODEL_HOME=<XMODEL_INSTALLATION_PATH>
export PATH=${PATH}:${XMODEL_HOME}/bin
export SALMD_LICENSE_FILE=9524@<LICENSE_HOSTNAME>
export XMODEL_SIMULATOR=<VERILOG_SIMULATOR>
```

where the fields **<XMODEL_INSTALLATION_PATH>** and **<LICENSE_HOSTNAME>** should be replaced with the full path to the **XMODEL** installation directory and license server hostname, respectively.

The field **<VERILOG_SIMULATOR>** can be either 'vcs', 'modelsim', or 'ncverilog'. If **\$XMODEL_SIMULATOR** environment variable is left undefined, the **XMODEL** simulator will try to find the simulator installed on your system automatically.

For csh-like shells, you can use the following commands (see **\$XMODEL_HOME/etc/DOT.cshrc** for example):

```
setenv XMODEL_HOME      <XMODEL_INSTALLATION_PATH>
setenv PATH              ${PATH}:${XMODEL_HOME}/bin
setenv SALMD_LICENSE_FILE 9524@<LICENSE_HOSTNAME>
setenv XMODEL_SIMULATOR <VERILOG_SIMULATOR>
```

Again, the fields **<XMODEL_INSTALLATION_PATH>** and **<LICENSE_HOSTNAME>** should be replaced with the full path to the **XMODEL** installation directory and license server hostname, respectively. The field **<VERILOG_SIMULATOR>** can be either 'vcs', 'modelsim', or 'ncverilog'.

It is convenient to set these environment variables within the shell start-up scripts (e.g. **~/ .bashrc** for bash-like shells and **~/ .cshrc** for csh-like shells) so that they can be properly set every time a new shell is started.

Also, since **XMODEL** runs on top of a Verilog simulator, it is necessary to set the proper environment variables for the Verilog simulator of your choice. For instance, for Synopsys® VCS®, one needs to set the VCS® installation directory (**\$VCS_HOME**), prepend its executable directory (**\$VCS_HOME/bin**) to the **\$PATH** variable, and define the Synopsys® license path (**\$SNPSLMD_LICENSE_FILE**). Please refer to the user manuals of your Verilog simulator for more details.

XMODEL_LICENSE_WAIT Environment Variable (Optional)

By default, an application program aborts immediately if it cannot check out the required license features. For instance, a feature '**XMODEL**' is required for running **XMODEL** simulation. Instead of aborting immediately, you can make the program wait for the next available license for a specified time duration (called '*license queuing*') by defining the **\$XMODEL_LICENSE_WAIT** environment variable.

The basic usage of this **\$XMODEL_LICENSE_WAIT** environment variable is to enable or disable license queuing. For bash-like shells, you can use the following commands to enable or disable license queuing, respectively:

```
export XMODEL_LICENSE_WAIT=yes      # to enable license queuing
export XMODEL_LICENSE_WAIT=no       # to disable license queuing
```

And for csh-like shells, you can use the following commands:

```
setenv XMODEL_LICENSE_WAIT yes      # to enable license queuing
setenv XMODEL_LICENSE_WAIT no       # to disable license queuing
```

A more advanced usage of the **\$XMODEL_LICENSE_WAIT** environment variable is to specify the time period for which the application waits for the next available license. You can define this wait period in units of minutes by assigning an integer value to the **\$XMODEL_LICENSE_WAIT** environment variable. For instance, the following command enables the license queuing and sets the wait period to be 20-minutes for bash-like shells:

```
export XMODEL_LICENSE_WAIT=20
```

The equivalent command in csh-like shells is:

```
setenv XMODEL_LICENSE_WAIT 20
```

Note that setting the wait period to zero is equivalent to disabling license queuing. If you enable license queuing by setting **\$XMODEL_LICENSE_WAIT** to **yes**, the wait period would be infinite.

3.3 Configuring *GLISTER* on Cadence® Virtuoso®

If you'd like to use **GLISTER** to graphically compose the analog/mixed-signal models on Cadence® Virtuoso® Schematic Editor and netlist them into **XMODEL** source codes, you need to perform the following additional steps.

Installing **XMODEL** Registry Files

GLISTER introduces new viewtypes such as "**xmodel**" and "**tb_xmodel**" as well as new environment variables and menu interfaces for the *GLISTER-ADE* integration. To put them into effect, you need to install the **XMODEL** registry files. There are multiple ways of doing this depending on whether you're a CAD administrator or an end-user.

If you are a CAD administrator with the write permission into the Cadence® installation directory and you'd like the **XMODEL** registry files to be effective for all projects using a particular version of Cadence® Virtuoso®, the easiest way is to use the following command to install the **XMODEL** registry files directly into the Cadence® Virtuoso® installation directory:

```
$XMODEL_HOME/bin/setup_registry --cdshome $CDSHOME
```

Here, the **\$XMODEL_HOME** is the environment variable defining the **XMODEL** installation path and the **\$CDSHOME** is the environment variable defining the installation path of the Cadence® IC package. The older versions of Cadence® IC package (e.g. IC5.1) may use **\$CDS_INST_DIR** instead of **\$CDSHOME**.

An example of running this command is shown below. You may supply an additional **--dryrun** option to the registry installer script **setup_registry** to see what actions the script will perform before actually installing the registry files.

```
$ $XMODEL_HOME/bin/setup_registry --cdshome $CDSHOME
*** setup_registry (XMODEL Release 2025.02 (x86_64))
The XMODEL registry files will be installed into the following Cadence installation
directory:
    /cad/cadence/IC_06.17.717

Do you want to proceed? [y/N]: y

Installing the XMODEL registry files...
cp -R /cad/scianalog/xmodel_2025.02/cadence/etc/registry/*
/cad/cadence/IC_06.17.717/share/cdssetup/registry/
cp -R /cad/scianalog/xmodel_2025.02/cadence/etc/tools/*
/cad/cadence/IC_06.17.717/tools/dfII/etc/tools/

XMODEL registry files are successfully installed.
```

As of **XMODEL** Release 2025.02, the registry installer script installs the following four files into the Cadence® installation directory:

- `$CDSHOME/share/cdssetup/registry/data/xmodel.reg`
- `$CDSHOME/share/cdssetup/registry/tools/xmodel.reg`
- `$CDSHOME/tools/dfll/etc/tools/xmodel/xmodel.cdseenv`
- `$CDSHOME/tools/dfll/etc/tools/menus/xmodel.menus`

Note that these files do not interfere with any other parts of the Cadence® software or user customizations. Hence, it is safe to install these files without worrying about affecting your existing customizations.

If you don't have the administrator privilege to modify the Cadence® installation directory, or if you want the **XMODEL** registry files to be effective only for a certain project or site, an alternative way is to install the **XMODEL** registry files into one of the local directories that can be reached by the Cadence® Setup Search File (CSF) mechanism. The local directory can be either your local work directory where you launch Cadence® Virtuoso® from, your home directory (`$HOME`), project directory (`$CDS_PROJECT`), or site directory (`$CDS_SITE`).

Depending on where you install the registry files, the **XMODEL** registry files will take effects for different range of projects and users. For instance, if you install the registry files into a local work directory, they will be effective only for the Virtuoso sessions started from that directory. On the other hand, if you install them to your home directory, they will be effective for all Virtuoso sessions started by you. If you install them into the directory pointed by `$CDS_PROJECT` or `$CDS_SITE` environment variable, they will be effective to everyone who shares the same `$CDS_PROJECT` or `$CDS_SITE` setting.

You can use the following command to install the **XMODEL** registry files into a local directory of your choice. Basically, change into the local directory and execute the **setup_registry** script with a **--local** option as below:

```
cd $CDS_PROJECT or <your cadence work directory>
$XMODEL_HOME/bin/setup_registry --local .
```

An example of running this command is shown below. Again, you may use an additional **--dryrun** option to see what actions the registry installer script will perform before actually installing the registry files.

```
$ cd /home/user/cadence
$ $XMODEL_HOME/bin/setup_registry --local .
*** setup_registry (XMODEL Release 2025.02 (x86_64))
The XMODEL registry files will be installed into the following local directory:
    /home/user/cadence

Do you want to proceed? [y/N]: y

Installing the XMODEL registry files...
mkdir -p /home/user/cadence/menus
cp -RL /cad/scianalog/xmodel_2025.02/cadence/etc/menus/* /home/user/cadence/menus/
cp -L /cad/scianalog/xmodel_2025.02/cadence/etc/xmodel.cdsenv /home/user/cadence/
cp /cad/scianalog/xmodel_2025.02/cadence/etc/data.reg /home/user/cadence/data.reg
cp /cad/scianalog/xmodel_2025.02/cadence/etc/csfLookupConfig
/home/user/cadence/csfLookupConfig

XMODEL registry files are successfully installed.
```

As of **XMODEL** Release 2025.02, the registry installer script installs or modifies the following four files in the local directory:

- xmodel.cdsenv
- menus/xmodel.menus
- data.reg
- csfLookupConfig

For more information about the CSF mechanism or the **\$CDS_PROJECT** and **\$CDS_SITE** directories, please refer to the *Cadence® Application Infrastructure User Guide* located in **\$CDSHOME/doc/caiuser/caiuser.pdf**.

Editing .cdsinit File

The next step is to configure your Cadence® Virtuoso® environment so that the Virtuoso® Schematic Editor executes the following SKILL commands upon start-up:

```
; XMODEL/Virtuoso Integration Setup
XMODEL_HOME = getShellEnvVar("XMODEL_HOME")
loadContext( strcat( XMODEL_HOME "/cadence/etc/xmodel.cxt" ) )
callInitProc( "xmodel" )
```

For instance, you can insert these lines in the site-wide or personal **.cdsinit** file which is located in the **\$CDS_SITE** directory or in the Cadence® working directory (**\$CDS_PROJECT**), respectively. Note that the shell environment variable **\$XMODEL_HOME** must be defined prior to launching Virtuoso®.

Editing cds.lib File

To access the libraries of the **XMODEL** primitive symbols and modeling examples, you need to edit your **cds.lib** file to include the **XMODEL**-related libraries. Please refer to **\${XMODEL_HOME}/cadence/etc/cds.lib** for the lines to be inserted in your **cds.lib**.

One way is to simply include this whole **cds.lib** file into yours by adding the following line in your local **cds.lib** file:

```
INCLUDE ${XMODEL_HOME}/cadence/etc/cds.lib
```

Or, you can selectively include the libraries by adding the lines individually:

```
DEFINE xmodel_prims ${XMODEL_HOME}/cadence/xmodel_prims  
DEFINE xmodel_blocks ${XMODEL_HOME}/cadence/xmodel_blocks
```

The '**xmodel_prims**' and '**xmodel_blocks**' libraries are the minimal set of libraries required to use **GLISTER**.

XMODEL_SIMDIR Environment Variable (Optional)

When **GLISTER** performs netlisting, it stores the generated model files and other supporting files in the directory defined by the **\$XMODEL_SIMDIR** environment variable.

In a bash-like shell, you can define this environment variable by using the following shell command:

```
export XMODEL_SIMDIR=$CDS_PROJECT/xmodel.sim
```

And in a csh-like shell, you can define it by:

```
setenv XMODEL_SIMDIR $CDS_PROJECT/xmodel.sim
```

If the **\$XMODEL_SIMDIR** is not defined, the assumed default value is **\$CDS_PROJECT/xmodel.sim**. If **\$CDS_PROJECT** is not defined, then **\$XMODEL_SIMDIR** is defaulted to **\$HOME/xmodel.sim**.

XMODEL_LICENSE_TIMEOUT Environment Variable (Optional)

Using **GLISTER** features within the Cadence® Virtuoso® Design Environment checks out a **GLISTER** license feature. The checked-out **GLISTER** license does not get checked back in unless the user explicitly does so (e.g. by closing the Virtuoso session) or a specified inactivity period elapses.

You can customize this time-out period by defining the **\$XMODEL_LICENSE_TIMEOUT** environment variable. For instance, using the following shell command in a bash-like shell defines the time-out period as 240 minutes (i.e., 4 hours):

```
export XMODEL_LICENSE_TIMEOUT=240
```

The equivalent command in a csh-like shell is:

```
setenv XMODEL_LICENSE_TIMEOUT 240
```

If you prefer setting the time-out period in the **.cdsinit** file, you can define a SKILL variable named **xmodelLicenseTimeOut** with the desired time-out period in minutes *after* the **XMODEL/GLISTER** context file is loaded:

```
; XMODEL/Virtuoso Integration Setup
XMODEL_HOME = getShellEnvVar("XMODEL_HOME")
loadContext( strcat( XMODEL_HOME "/cadence/etc/xmodel.cxt" ) )
callInitProc( "xmodel" )

xmodelLicenseTimeOut = 240
```

If neither the **\$XMODEL_LICENSE_TIMEOUT** environment variable nor the **xmodelLicenseTimeOut** SKILL variable is defined, the default time-out period of 24 hours (i.e., 1440 minutes) is assumed.

3.4 Additional Configuration for **MODELZEN**

If you wish to use **MODELZEN** from Cadence® Virtuoso® Schematic Editor using the **GLISTER** interface, you may need to perform some additional setups:

- Override **MODELZEN**-related **GLISTER** default options if necessary
- Configure how **MODELZEN** extracts circuit netlists from schematic cellviews (`modelzenGenNetlist()`)

Overriding **MODELZEN**-related **GLISTER** Default Options

GLISTER provides some **MODELZEN**-related options that control the ways **GLISTER** executes **MODELZEN** within Cadence® Virtuoso®. These options can be modified from the *Set MODELZEN Options* dialog box that opens when selecting the *GLISTER→Set MODELZEN Options* menu from the Cadence® Virtuoso® Schematic Editor window.

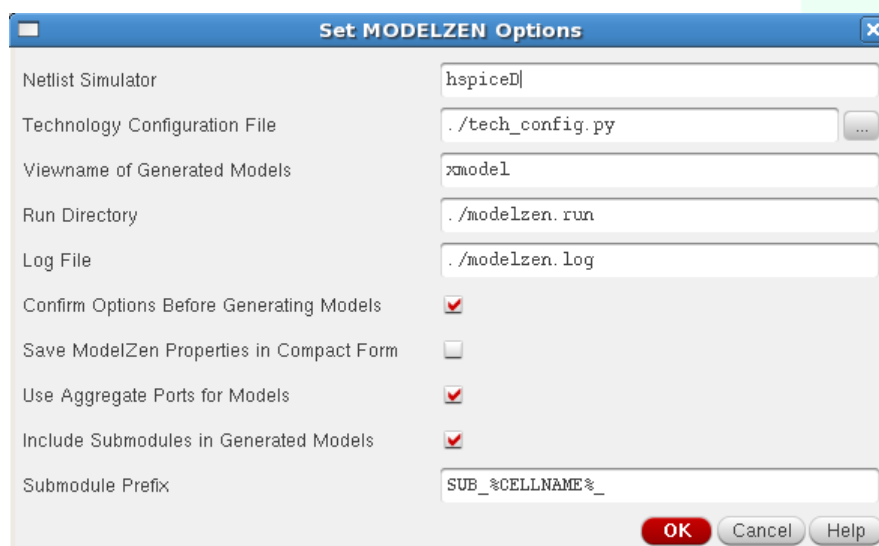


Figure 1. The dialog box for setting **MODELZEN options.**

For instance, the *Netlist Simulator* field sets the format of the circuit netlist to be extracted from the schematic cellview and the *Technology Configuration File* field sets the default technology configuration file to be used by **MODELZEN**. Please refer to the *MODELZEN User's Guide* for more detailed descriptions on each of these options.

One can override the default values of the **MODELZEN**-related **GLISTER** options by overriding the values of the SKILL variables that correspond to these options after loading the **XMODEL/GLISTER** context file. An excerpt of the Cadence® Virtuoso® initialization file `.cdsinit` that overrides some of the **MODELZEN**-related **GLISTER** options is listed below. The first part initializes the **XMODEL/GLISTER** interface for Cadence® Virtuoso® as previously explained in [Section 3.3](#). The second part redefines some SKILL variables with a prefix "`modelzen`" to override the default values of the **MODELZEN**-related **GLISTER** options.

```
; XMODEL/Virtuoso Integration Setup
printf( "Loading XMODEL/Virtuoso Integration...\n" )
XMODEL_HOME = getShellEnvVar("XMODEL_HOME")
loadContext( strcat( XMODEL_HOME "/cadence/etc/xmodel.cxt" ) )
callInitProc( "xmodel" )

; MODELZEN options (NOTE: must come after context file loading)
modelzenDefaultModelView = "xmodel"
modelzenDefaultConfigFile = "./tech_config.py"
modelzenDefaultRunDir = "./modelzen.run"
modelzenDefaultNetlistSimulator = "hspiceD"
modelzenSubModulePrefix = "SUB_%CELLNAME%"
modelzenShowOptionDialog = t
```

Table 2 lists the names of the SKILL variables that define the default values of the *MODELZEN*-related *GLISTER* options shown in Figure 1.

Table 2. Listing of the SKILL variables setting the default values of the *MODELZEN*-related *GLISTER* options.

Option Name	Corresponding SKILL Variable Name
<i>Netlist Simulator</i>	<code>modelzenDefaultNetlistSimulator</code>
<i>Technology Configuration File</i>	<code>modelzenDefaultConfigFile</code>
<i>Viewname of Generated Models</i>	<code>modelzenDefaultModelView</code>
<i>Run Directory</i>	<code>modelzenDefaultRunDir</code>
<i>Log File</i>	<code>modelzenDefaultLogFile</code>
<i>Confirm Options Before Generating Models</i>	Set <code>modelzenShowOptionDialog</code> as <code>t</code> to select this flag
<i>Save MODELZEN Properties in Compact Form</i>	Set <code>modelzenIncludeAllPinInfo</code> as <code>nil</code> to select this flag
<i>Use Aggregate Ports for Models</i>	Set <code>modelzenUseAggregatePorts</code> as <code>t</code> to select this flag.
<i>Include Submodules in Generated Models</i>	Set <code>modelzenSubModulePrefix</code> with any string to select this flag; setting it as <code>nil</code> unselects this flag.
<i>Submodule Prefix</i>	<code>modelzenSubModulePrefix</code>

Configure How *MODELZEN* Extracts Circuit Netlists from Schematic Cellviews

Running ***MODELZEN*** using the ***GLISTER*** interface involves extracting a circuit netlist from the schematic cellview. If your design environment uses a customized procedure or interface for extracting SPICE or Spectre netlists from the schematic cellview, you may need to redefine a SKILL procedure named

modelzenGenNetlist() in order for **MODELZEN** to use the same steps when extracting the circuit netlists.

The usage format of the SKILL function **modelzenGenNetlist()** is shown as below:

```
modelzenGenNetlist( libName cellName viewName simulator )
```

The function takes four arguments **libName**, **cellName**, **viewName**, and **simulator** where **libName**, **cellName**, and **viewName** are the library, cell, and view names of the design cellview to be netlisted and **simulator** is the name of the Cadence® netlist simulator that determines the format of the netlist (e.g. **hspiceD**, **spectre**, and **cd1**). The function then extracts a netlist from the cellview and returns the full path to the extracted netlist file.

To redefine the **modelzenGenNetlist()** function, one may need to know the basics of SKILL programming and specific steps required for netlisting in her or his design environment. Please refer to the appropriate Cadence® documentations, or consult your PDK providers or CAD administrators in case you need further help.

The following codes show the default **modelzenGenNetlist()** function which is defined when the **XMODEL/GLISTER** context file is first loaded. It extracts the netlist using the Cadence® Analog Design Environment (ADE) SKILL functions such as **simulator()**, **design()**, and **createNetlist()**. If you can extract netlists from your schematic cellview using the standard ADE interface, it is likely that you do not need to redefine **modelzenGenNetlist()**.

```

; modelzenGenNetlist( libName cellName viewName simulator ):
; generates the 'simulator' netlist of a cellview and
; returns the full filepath of the generated netlist
procedure( modelzenGenNetlist( libName cellName viewName simulator )
  let( ( netlistFile netlistDir )
    ; initialize simulator
    simulator( simulator )

    ; generate netlist
    design( libName cellName viewName )
    unless( netlistFile = createNetlist( ?recreateAll t ?display nil )
      error( "error occurred while netlisting %s.%s:%s."
        libName cellName viewName )
    )
    ocnCloseSession()

    ; return netlist filename
    netlistDir = xmodelGetDirName( netlistFile )
    strcat( netlistDir "/netlist" )
  )
)

```

3.5 Testing with Examples

To check whether the installation has been done successfully, use the following commands to execute a testbench for a charge-pump PLL example. For more information on the charge-pump PLL model itself and detailed description on the testbenches, please refer to the documentation titled *Simulating Charge-Pump PLL in **XMODEL***.

```

unix> mkdir ~/xmodel
unix> cp -R $XMODEL_HOME/example ~/xmodel
unix> cd ~/xmodel/example/cp_pll/tb/tb_locking
unix> make tb_locking
unix> xwave -a xmodel.jez

```

The execution is successful if the simulation is run without errors and the last command pops-up a window displaying waveforms as shown below.

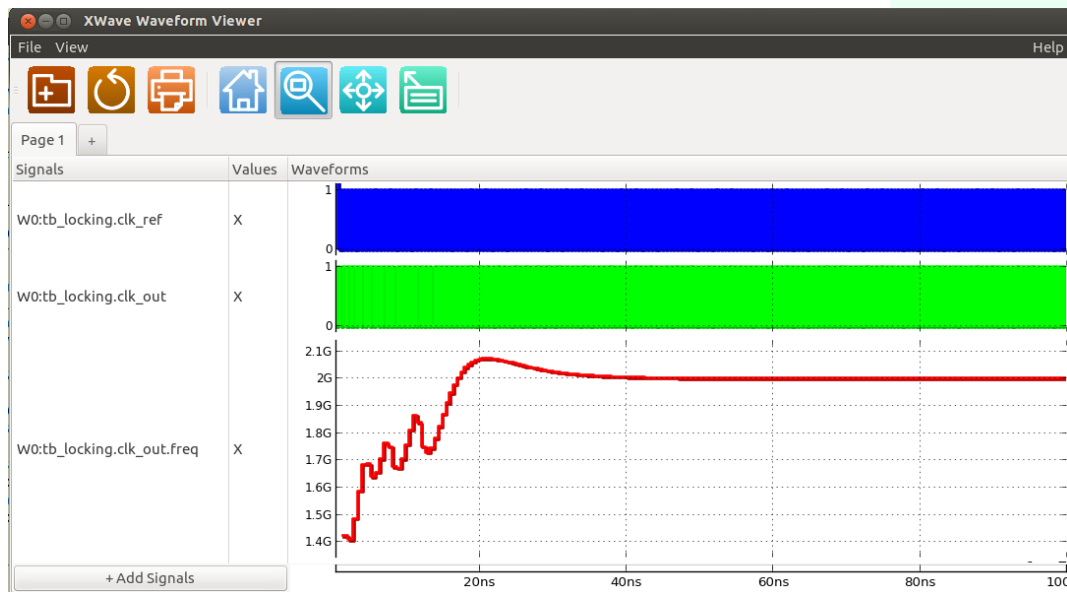


Figure 2. Locking transient simulation of a charge-pump phase-locked loop.

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